

Interaction-driven valley polarization at a higher-order Van Hove singularity: a Variational Quantum Eigensolver benchmark

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October 2, 2026

Abstract

We study the effect of a Hubbard interaction on a higher-order Van Hove singularity (HOVHS) in the Chern band of a topological-insulator moiré surface state. Projecting onto the two inequivalent moiré valleys K and K' , we solve the half-filled interacting problem by exact diagonalization and find that the interaction drives spontaneous valley-Ising order: the ground state polarizes into a single valley, protected as a two-fold degenerate doublet by a $U(1)$ valley symmetry. The order onsets at arbitrarily weak coupling, consistent with the divergent density of states at the HOVHS, and is independent of the sublattice polarization of the single-particle states. The two-valley model is deliberately small so that exact diagonalization provides ground truth; we use it to benchmark a variational quantum eigensolver (VQE), establishing a validated pipeline that can be scaled to larger patches where exact diagonalization becomes infeasible.

1 Introduction

Van Hove singularities, where the electronic density of states (DOS) diverges, are focal points for interaction-driven order: the enhanced DOS amplifies the effect of electron–electron interactions, so that even weak coupling can trigger magnetic, charge-ordered, or superconducting instabilities. A *higher-order* Van Hove singularity (HOVHS), at which the dispersion is flatter than an ordinary saddle point and the DOS diverges as a power law rather than logarithmically, intensifies this mechanism and makes the weak-coupling instability especially sharp.

Recent work established that the moiré surface state of a three-dimensional topological insulator, described by a single Dirac cone subject to a C_6 -symmetric moiré potential and a Zeeman field, hosts a pair of cubic HOVHS at the $\pm K$ corners of the moiré Brillouin zone within a Chern band [1]. That study is purely single-particle. The natural and physically central question it leaves open is what interactions do to such a singularity: a Dirac system with a divergent DOS at the Fermi level is a prototypical setting for correlated order, and the specific instability that wins, magnetic, valley, charge, or pairing, is not determined by the band structure alone.

This project addresses that question from two complementary perspectives. The first is condensed-matter: we add an on-site Hubbard repulsion to the HOVHS manifold and determine, nonperturbatively, the nature of the interaction-driven order. The second is methodological: solving correlated electron problems is a natural target for quantum algorithms, and the variational quantum eigensolver (VQE) is a leading near-term candidate. To trust VQE on a correlated problem one must first validate it against exact results on a system small enough to diagonalize exactly. The two-valley (K , K') model studied here is constructed precisely as such a benchmark: it is the minimal setup that retains the essential intervalley scattering physics of the HOVHS while remaining exactly solvable. Establishing that VQE reproduces the exact ground state here provides a validated pipeline

that can subsequently be scaled up—by including additional patch points and bands—to regimes where exact diagonalization is no longer feasible and the quantum algorithm becomes genuinely necessary.

2 Method

Single-particle model. We begin from the surface-state Hamiltonian

$$H_0 = v_F \hat{\mathbf{z}} \cdot (\mathbf{k} \times \boldsymbol{\sigma}) + V(\mathbf{r}) \sigma_0 + h_z \sigma_z, \quad (1)$$

with a C_6 -symmetric moiré potential $V(\mathbf{r})$. Expanding in plane waves and diagonalizing on a truncated reciprocal-lattice grid yields the moiré bands; for the chosen parameters a cubic HOVHS sits at the $\pm K$ valleys, with the two valleys degenerate by C_6 symmetry.

Two-valley projection. Because the singular DOS is concentrated at $\pm K$, we project onto a patch consisting of the two inequivalent corners K and K' . On the hexagonal moiré lattice these two momenta form a set closed under momentum addition modulo reciprocal lattice vectors (using $3\mathbf{K} \equiv 0$), so intervalley scattering is captured exactly and umklapp processes fold back onto the patch. Retaining a small set of bands near the singularity gives a compact set of fermionic modes labeled by valley and band.

Interaction. We add an on-site Hubbard repulsion and express it in the band basis through the spin-projected Bloch overlap form factors. To keep the construction manifestly Hermitian and free of operator-ordering ambiguities, we build the interaction from band-projected density operators,

$$H_{\text{int}} = \frac{U}{N_k} \sum_{\mathbf{q}} \rho_{\uparrow}(\mathbf{q}) \rho_{\downarrow}(-\mathbf{q}), \quad (2)$$

whose Hermiticity follows from $\rho_{\sigma}(\mathbf{q})^{\dagger} = \rho_{\sigma}(-\mathbf{q})$ and which generates the required normal-ordering one-body correction automatically.

Filling and diagonalization. The relevant physics lives in the valley-degenerate manifold at the HOVHS energy E_K . We work at the filling that places the Fermi level in this manifold—half-filling of the two-fold valley manifold—and diagonalize the interacting Hamiltonian in the corresponding fixed-particle-number sector, where particle-number conservation makes the sector exact and small. At $U = 0$ this sector has a two-fold degenerate ground state corresponding to the two ways of occupying the valley manifold.

Characterizing the order. We treat the valley label as a pseudospin and measure its three components in the ground space: the valley polarization $T_z = N_K - N_{K'}$ (diagonal) and the intervalley coherences T_x, T_y (off-diagonal). Projecting these operators into the degenerate ground space gives a basis-independent readout, and the commutators $[H, T_a]$ identify which components are conserved, distinguishing Ising from XY order.

VQE benchmark. Finally, we map the same Hamiltonian to qubits via the Jordan–Wigner transformation and run a particle-number-constrained VQE, comparing its ground state against the exact-diagonalization result to validate the method on this small, exactly solvable system.

3 Results

Valley-Ising order. Turning on the Hubbard interaction at half-filling of the HOVHS manifold drives the system into a fully valley-polarized state: the ground-space valley polarization takes the extremal values $T_z = \pm 1$, so the extra manifold charge resides entirely in one valley. The two polarized states remain exactly degenerate for all U , forming a symmetry-protected doublet—the finite-size signature of spontaneous symmetry breaking, with the two members corresponding to the two valley orientations related by the $K \leftrightarrow K'$ symmetry. The interacting spectrum (Fig. 1) shows the whole sector reorganizing while every level, including the ground doublet, retains its exact two-fold degeneracy.

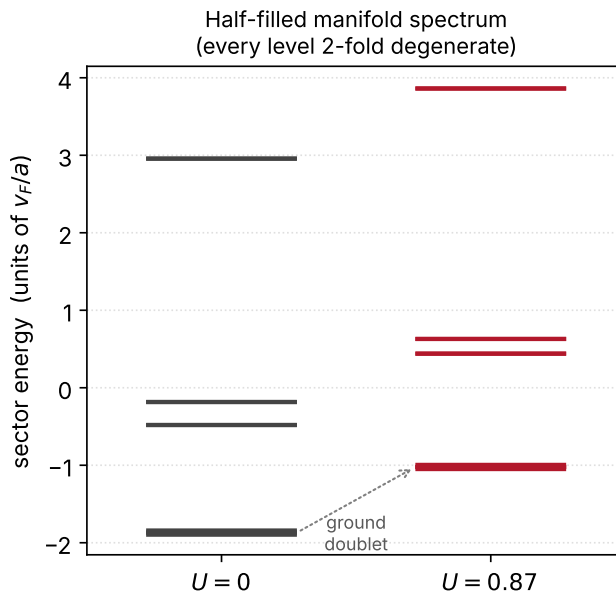


Figure 1: Lowest eigenvalues of the half-filled two-valley sector at $U = 0$ (left) and $U = 0.87$ (right). Every level is exactly two-fold degenerate. The interaction shifts and reorganizes the spectrum, yet the ground-state doublet persists, reflecting the $U(1)$ valley symmetry that protects the valley-polarized pair.

The order is specifically *Ising*, not *XY*. The Hamiltonian conserves the valley polarization, $[H, T_z] = 0$ (a $U(1)$ valley symmetry), but not the intervalley coherences, $[H, T_x], [H, T_y] \neq 0$. The valley pseudospin is therefore pinned to the polarization axis: intervalley-coherent superpositions are not ground states, and the broken symmetry is the discrete $K \leftrightarrow K'$ exchange. We emphasize that the extremal eigenvalues of T_x, T_y obtained by projecting into a two-state ground space are a generic feature of any pseudospin on a two-level system and do *not* indicate coherence order; the commutation structure, not the projected eigenvalues, fixes the universality class.

Weak-coupling onset. The valley-polarization order parameter lifts off zero as soon as U is turned on, with no finite threshold (Fig. 2). This is the expected hallmark of a HOVHS-driven instability: the power-law divergent DOS removes the kinetic cost of polarizing the nearly flat manifold, so interactions order the system at arbitrarily weak coupling, in the spirit of a Stoner instability carried by the valley degree of freedom.

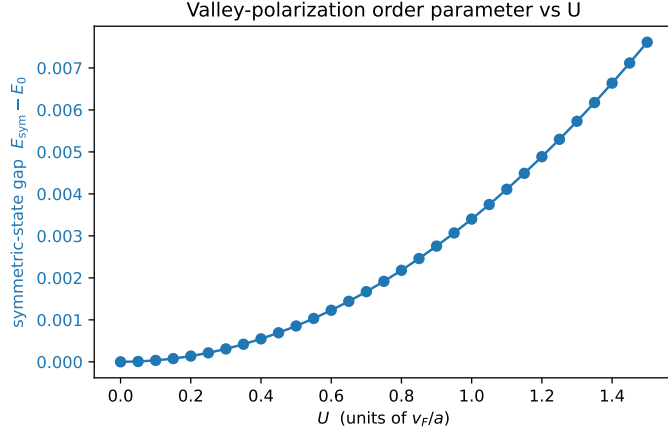


Figure 2: Valley-polarization order parameter, measured as the energy gap $E_{\text{sym}} - E_0$ between the valley-symmetric state and the ground state, as a function of the Hubbard interaction strength U (units of v_F/a). The gap rises from zero immediately as U is turned on, with no finite critical coupling: the divergent density of states at the HOVHS drives valley-polarized order at arbitrarily weak interaction.

Independence from sublattice polarization. The single-particle HOVHS states are themselves sublattice/spin-polarized, a consequence of the broken time-reversal symmetry, and one might ask whether this single-particle polarization is responsible for the valley order. It is not. Sweeping the Zeeman field along the HOVHS line—holding the singularity fixed while varying the wavefunction structure—the sublattice polarization and the corresponding imbalance of the intervalley form factors grow substantially, whereas the valley-polarization instability remains essentially unchanged and is already fully developed at zero Zeeman field, where the sublattice polarization vanishes (Fig. 3). The valley-Ising order is thus driven by the intervalley Hubbard scattering and the divergent DOS, not by the sublattice character of the single-particle states. This cleanly separates the many-body instability from the single-particle physics of the underlying band.

VQE validation. The particle-number-constrained VQE reproduces the exact-diagonalization ground state of the two-valley model, confirming that the variational method captures the correlated valley-polarized state. Because the benchmark system is exactly solvable, this establishes the fidelity of the pipeline directly, calibrating the ansatz and optimizer for subsequent application to larger patches where exact diagonalization is infeasible.

4 Outlook

The two-valley model establishes both a physical result, HOVHS-driven valley-Ising order, independent of sublattice polarization, and a validated quantum-classical pipeline. The natural continuation is to enlarge the patch beyond the two corners, incorporating additional momenta and bands to resolve competing instabilities and finite-size effects, in the regime where the VQE approach moves from a benchmarked method to a necessary tool.

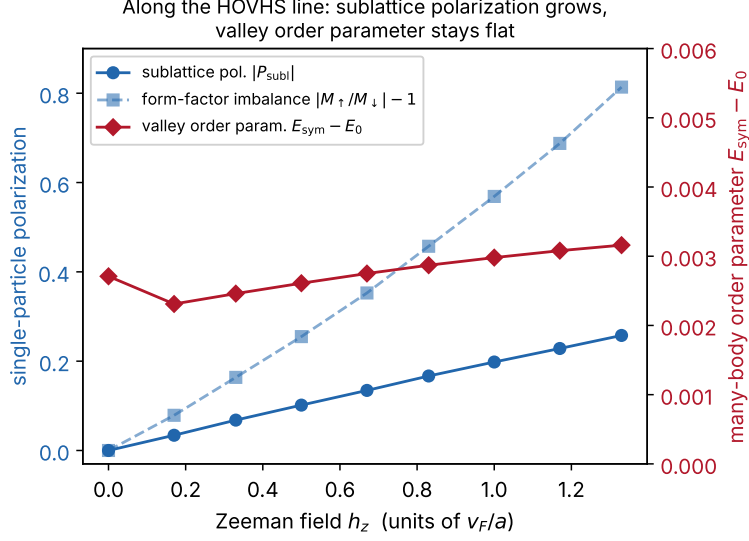


Figure 3: Sweep along the HOVHS line. As the Zeeman field h_z increases, the single-particle sublattice polarization and the intervalley form-factor imbalance (blue, left axis) grow monotonically, while the many-body valley-polarization order parameter $E_{\text{sym}} - E_0$ (red, right axis) stays essentially flat and is already finite at $h_z = 0$, where the sublattice polarization is zero. The instability does not track the sublattice polarization.

References

- [1] L. Pullasserri and L. H. Santos, *Chern Bands with Higher-Order Van Hove Singularities on Topological Moiré Surface States*, arXiv:2402.16772.